

Multi-Robot Path Optimization for Search and Rescue

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Outline

Problem Overview

Assumptions

Solution Representation

The Objective Function

Constraints

Simulation Video

Problem Overview

Objective: Optimize paths for multiple robots to explore an unknown area.

Key Challenges:

- ▶ Maximize area coverage.
- ▶ Maintain communication between robots.
- ▶ Adhere to energy (battery) limitations.
- ▶ Avoid obstacles and collisions.
- ▶ Handle network disconnections gracefully.

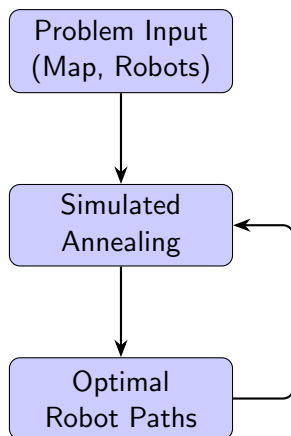
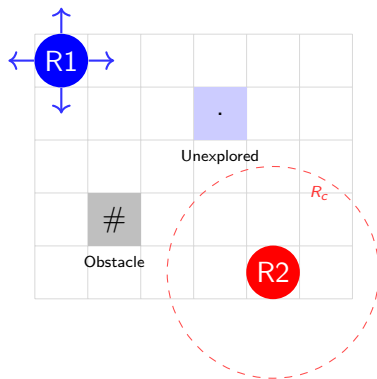


Figure: High-level overview of the optimization process.

Assumptions at a Glance

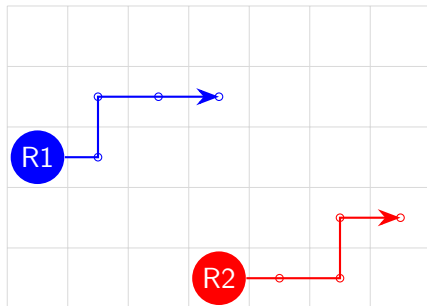
- ▶ **Environment:** Discrete grid map with unexplored cells, free space, and obstacles.
- ▶ **Robots:** Team of identical robots.
- ▶ **Obstacles:** Unknown initially, discovered during exploration.
- ▶ **Motion:** Discrete time steps, 4-connected grid (up, down, left, right, stay).
- ▶ **Energy:** Finite budget limits travel distance.
- ▶ **Communication:** Position sharing; link quality degrades with distance.



How a Solution Looks Like

A solution is a set of paths, one for each robot, over a fixed time horizon K .

Path Array: A 3D array $path_array[r][t] = (x, y)$ stores the position of robot r at time step t .



Example paths for 2 robots over 4 time steps.

Objective Function: The Goal

The goal is to **maximize** a composite objective function that balances multiple competing goals:

$$\begin{aligned} f = & \alpha \cdot \text{Coverage} \\ & + \beta \cdot \text{Connectivity} \\ & - \gamma \cdot \text{Disconnection Penalty} \end{aligned}$$

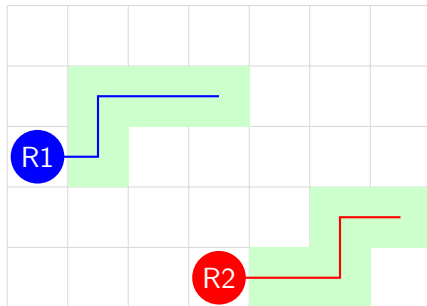
where α, β, γ are weights to tune the importance of each component.

Objective 1: Coverage

Goal: Maximize the number of unique unexplored cells visited by all robots.

$$\text{Coverage} = \sum_{\text{all cells } (x,y)} C_{x,y}$$

where $C_{x,y} = 1$ if a cell (x,y) is visited, and 0 otherwise.



Shaded cells are newly explored and contribute to the coverage score.

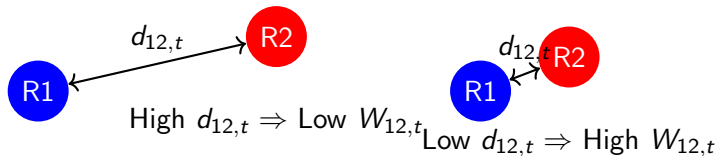
Objective 2: Connectivity

Goal: Encourage robots to stay close to each other to ensure reliable communication.

$$\text{Connectivity} = \sum_{t=0}^K \sum_{i < j} W_{ij,t}$$

The link quality $W_{ij,t}$ between robots i and j is higher when they are closer:

$$W_{ij,t} = \frac{1}{1 + d_{ij,t}} \quad (\text{if } d_{ij,t} \leq R_c)$$

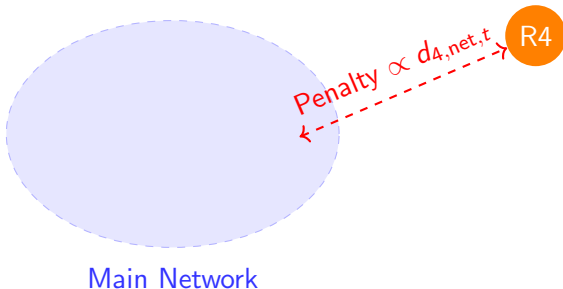


Penalty: Disconnection

Goal: Penalize robots that become disconnected from the main communication network.

$$P_{\text{disconnect}} = \sum_{t=0}^K \sum_{i=1}^N \delta_{i,t} \cdot d_{i,\text{net},t}$$

- ▶ $\delta_{i,t} = 1$ if robot i is disconnected at time t .
- ▶ $d_{i,\text{net},t}$ is the distance from the disconnected robot to the closest robot in the main network.



Hard Constraints (Feasibility)

A path is only valid if it satisfies all hard constraints. An infeasible path is given a very poor score (e.g., -1).

1. Map Bounds: Robots must stay within the grid.

$$0 \leq x_{i,t} < N, \quad 0 \leq y_{i,t} < M$$

2. Motion Model: Robots can only move to adjacent cells (4-way) or stay put.

$$|x_{i,t} - x_{i,t-1}| + |y_{i,t} - y_{i,t-1}| \leq 1$$

3. Energy Budget: Total distance traveled cannot exceed the battery limit ℓ .

$$\sum_{\tau=1}^t \text{dist}(p_{i,\tau}, p_{i,\tau-1}) \leq \ell$$

4. Obstacle Avoidance:

Robots cannot occupy cells with known obstacles.

$$p_{i,t} \notin \text{Obstacles}$$

5. Collision Avoidance: No two robots can be in the same cell at the same time.

$$p_{i,t} \neq p_{j,t} \quad \forall i \neq j$$

Live Simulation Demo